

NAVIGATING THE ZOO OF VISUALIZATION TECHNIQUES

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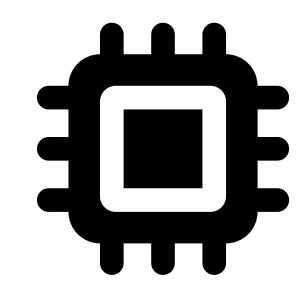
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Why

►What are **pathways through the space** of visualization techniques?

How

►**Study similarity** of visualization techniques for different types of data (see [1] for more details)



Model-Driven Approach

- Categorize vis. techniques based on Munzner's scheme [2]
- Define vis. technique signatures based on categorization
- Define similarity based on Jaro-Winkler distance between signatures

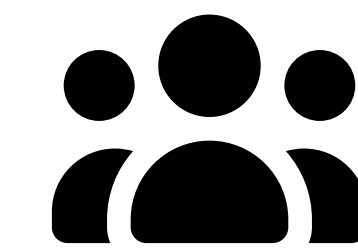
Marks:
Points (P), Lines (L),
Areas (A)

Channels:
Position (P), Length size (L),
Area size (A), Color (C), Shape (S)

DrDA MpML CpCACc RsOLLo
Example signature for Node-Link Diagram (NLD)

Data Facets:
Attributes (A), Space
(S), Time (T),
Relationship (R)

- Region: Separate (S), Order (O), Align (A)
- Axis Orientation: Rectilinear (L), Parallel (P), Radial (R)
- Layout: Dense (D), Space Filing (S)



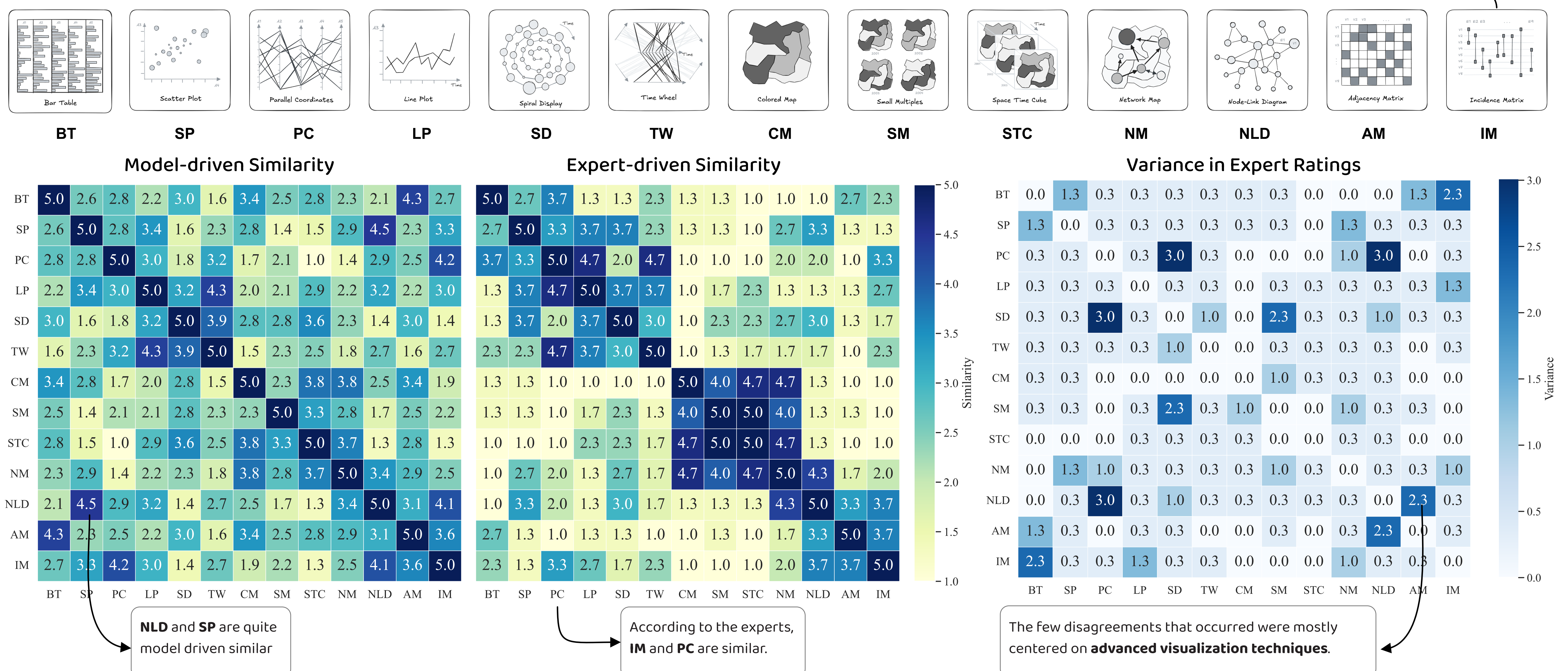
Expert-Driven Approach

- Set up online study with sketchy style vis. techniques
- Ask three experts for intuitive assessment of similarity
- Experts rate similarity of 78 pairs of vis. techniques

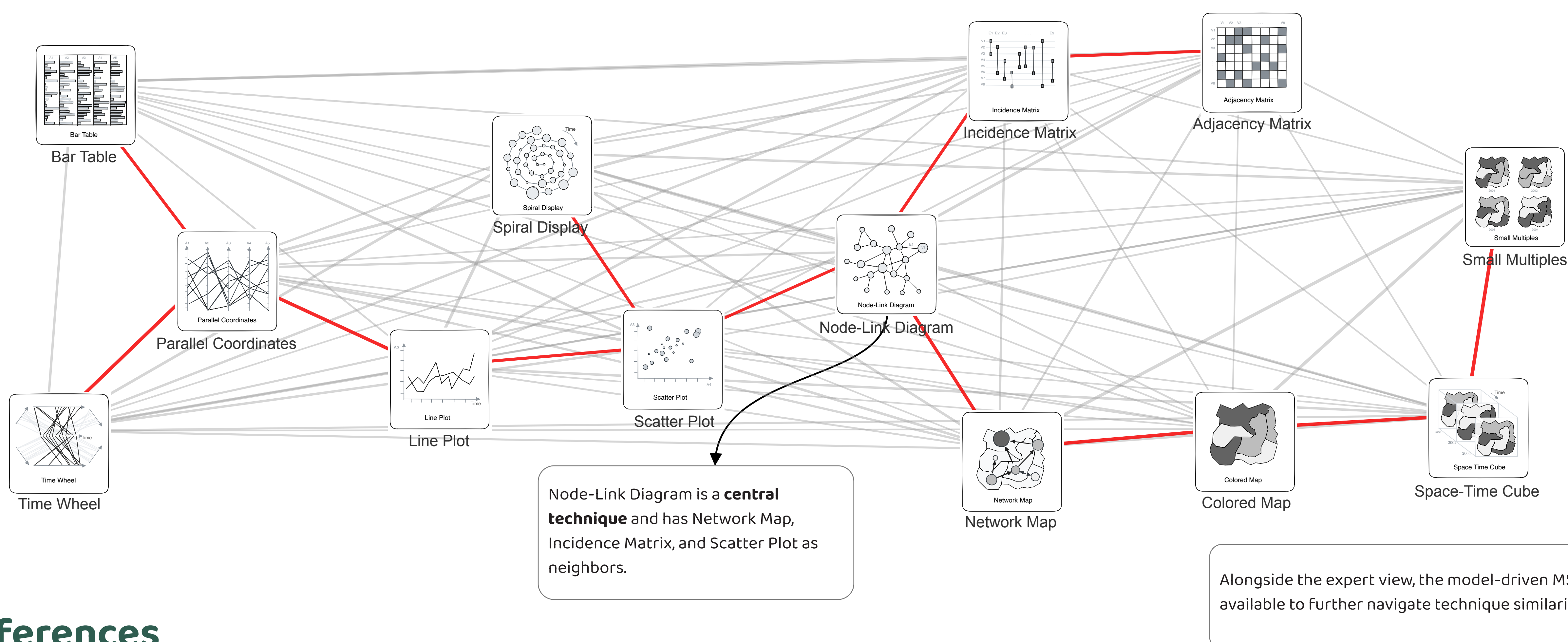
Results

►Heatmaps of model-driven and expert-driven **similarity of 13 visualization techniques**

Sketchy style vis. capturing the nature of the techniques rather than implementation details



►Similarity-based **minimum spanning tree (MST)** reveals **relationships** between vis. techniques



References

- [1] A. A. Salako and C. Tominski. Toward Understanding Similarity of Visualization Techniques. ArXiv, 2025. doi: 10.48550/arXiv.2506.17032
- [2] T. Munzner. Visualization Analysis and Design. A K Peters, 2014. doi: 10.1201/b17511